

Computer Network Project

Faculty of Computers and Artificial Intelligence (FCAI)

Project Supervision

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1. Introduction

This technical report documents the design, implementation, and configuration of a comprehensive enterprise network infrastructure spanning two main branches: the FCAI Branch (Educational/Labs) and the HQ Branch (Project Management & Administration). The primary objective is to establish a secure, scalable, and highly available network using Cisco devices.

The network utilizes the following technologies:

- VLANs for logical segmentation of traffic
- Router-on-a-Stick for inter-VLAN routing
- DHCP for automatic IP assignment
- DNS for domain name resolution
- OSPF protocol for dynamic routing between branches

2. Hardware Components Required

Component	Model	Role
Routers	Cisco 2911	Edge Gateways for FCAI and HQ branches
Multilayer Switch	Cisco 3560	Distribution Layer switches (HQ)
Access Switches	Cisco 2960	Connecting end-user devices to the network
Core Switch	Cisco 2960	Central hub for FCAI branch
DHCP Server	Generic Server	Hosting centralized DHCP services for FCAI
DNS Server	Generic Server	Resolving domain names to IP addresses

3. Network Design Strategy

- Logical Segmentation: 8 VLANs isolate traffic between labs (FCAI) and admin departments (HQ), reducing broadcast domains.
- Inter-VLAN Routing: Router-on-a-Stick with sub-interfaces and 802.1Q encapsulation for cross-VLAN communication.
- Dynamic Routing: OSPF (Area 0) for automatic route discovery across the Serial WAN link.
- IP Addressing (FLSM): Fixed Length Subnet Masking using /27 subnets (up to 30 hosts per VLAN).
- DHCP Services: Centralized DHCP server for FCAI with ip helper-address relay; HQ Router acts as DHCP server for HQ branch.
- Three-Tier Design (FCAI): Core, Distribution, and Access layers for scalability.
- Two-Tier Design (HQ): Collapsed Core and Access layers for simplicity.

4. VLAN and IP Network Design

Base Network: 192.168.1.0 | Subnet Mask: /27 (255.255.255.224) | Hosts per subnet: 30
WAN Link: 192.168.2.0/30 (255.255.255.252)

A. FCAI Branch (Educational Labs)

VLAN ID	Name	Network Address	Default Gateway	Broadcast
10	Lab3	192.168.1.0/27	192.168.1.1	192.168.1.31
20	TC	192.168.1.32/27	192.168.1.33	192.168.1.63
30	Lab1	192.168.1.64/27	192.168.1.65	192.168.1.95
40	Lab2	192.168.1.96/27	192.168.1.97	192.168.1.127

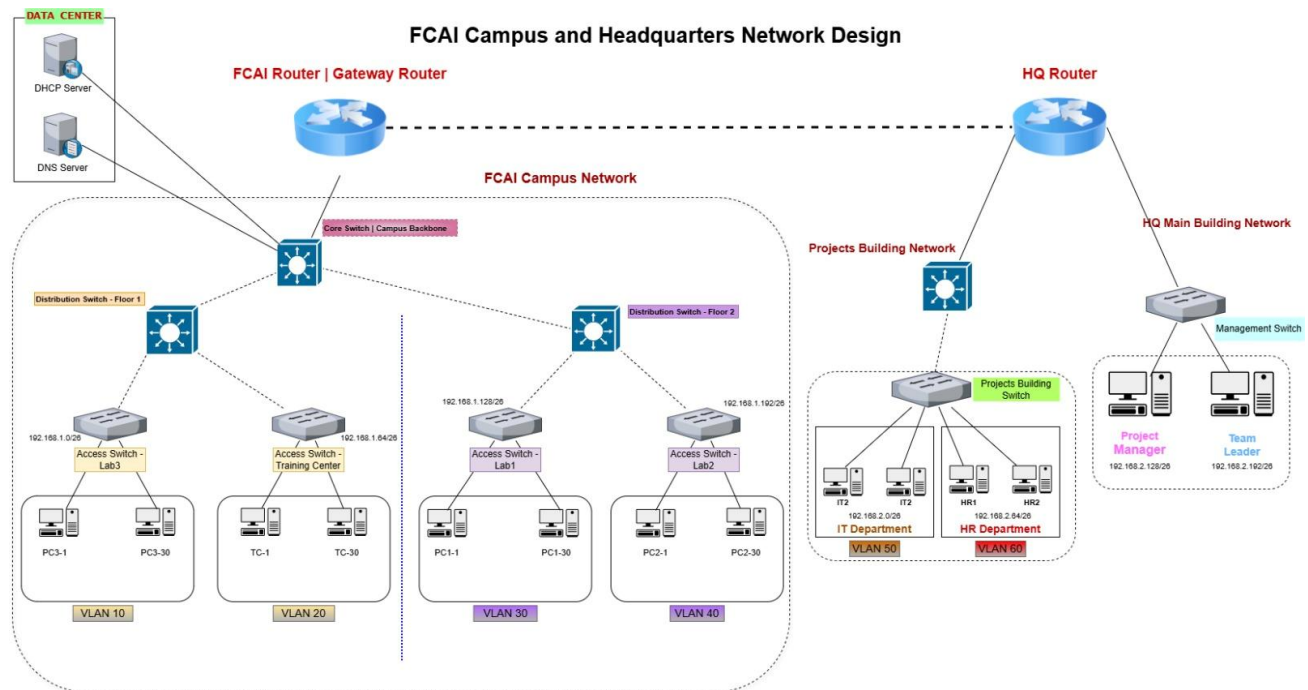
B. HQ Branch (Management)

VLAN ID	Name	Network Address	Default Gateway	Broadcast
50	IT	192.168.1.128/27	192.168.1.129	192.168.1.159
60	HR	192.168.1.160/27	192.168.1.161	192.168.1.191
70	Project Manager	192.168.1.192/27	192.168.1.193	192.168.1.223
80	Team Leader	192.168.1.224/27	192.168.1.225	192.168.1.255

C. WAN Link

Link	Network	FCAI Router IP	HQ Router IP
Serial Link	192.168.2.0/30	192.168.2.1	192.168.2.2

5. Network Topology Diagram Using Draw.IO



6. Network Diagram Explained

- FCAI Site: Three-tier hierarchical design. Core Switch serves as central hub connecting to the Data Center (DHCP + DNS servers) and two Distribution Layer switches. Each Distribution switch serves two Access switches for the labs.
- HQ Site: Two-tier design. Collapsed Core connects to HQ Router and distributes to Switch-IT-HR. Switch-PM-TL connects directly to HQ Router for VLANs 70 and 80.
- Trunking & Access: Router-to-switch and inter-switch links use 802.1Q Trunk links. End-user PC ports are configured as Access links.
- OSPF Routing: OSPF Area 0 on both routers enables dynamic route exchange across the Serial WAN link.

7. Network Configuration and Guidelines

Configurations are presented in order: FCAI infrastructure first, then HQ infrastructure, and finally the WAN/OSPF interconnection.

7.1 DHCP Server (192.168.1.2) — FCAI Branch

Static IP Configuration:

Parameter	Value
IP Address	192.168.1.2
Subnet Mask	255.255.255.224
Gateway	192.168.1.1

DHCP Pools:

Pool Name	Gateway	DNS Server	Start IP	Subnet Mask	Max Users
Lab3-Pool	192.168.1.1	192.168.1.3	192.168.1.4	255.255.255.224	28
TC-Pool	192.168.1.33	192.168.1.3	192.168.1.34	255.255.255.224	30
Lab1-Pool	192.168.1.65	192.168.1.3	192.168.1.66	255.255.255.224	30
Lab2-Pool	192.168.1.97	192.168.1.3	192.168.1.98	255.255.255.224	30

7.2 DNS Server (192.168.1.3) — FCAI Branch

Parameter	Value
IP Address	192.168.1.3

Parameter	Value
Subnet Mask	255.255.255.224
Gateway	192.168.1.1
DNS Service	ON

7.3 FCAI Router (Cisco 2911)

FCAI-Router — Router-on-a-Stick + OSPF

```
enable
configure terminal
hostname FCAI-Router
!
interface Serial10/3/0
 ip address 192.168.2.1 255.255.255.252
 clock rate 64000
 no shutdown
exit
!
interface GigabitEthernet0/0
 no shutdown
exit
!
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.1.1 255.255.255.224
exit
!
interface GigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 192.168.1.33 255.255.255.224
 ip helper-address 192.168.1.2
exit
!
interface GigabitEthernet0/0.30
 encapsulation dot1Q 30
 ip address 192.168.1.65 255.255.255.224
 ip helper-address 192.168.1.2
exit
!
interface GigabitEthernet0/0.40
 encapsulation dot1Q 40
 ip address 192.168.1.97 255.255.255.224
 ip helper-address 192.168.1.2
exit
!
router ospf 1
 network 192.168.1.0 0.0.0.255 area 0
 network 192.168.2.0 0.0.0.3 area 0
exit
!
do write
```

7.4 Core Switch (Cisco 2960)

Core-Switch — Central FCAI Hub

```
enable
configure terminal
hostname Core-Switch
!
vlan 10
  name Lab3
exit
vlan 20
  name TC
exit
vlan 30
  name Lab1
exit
vlan 40
  name Lab2
exit
!
interface FastEthernet0/5
  switchport mode access
  switchport access vlan 10
exit
!
interface FastEthernet0/1
  switchport trunk encapsulation dot1q
  switchport mode trunk
exit
!
interface FastEthernet0/2
  switchport trunk encapsulation dot1q
  switchport mode trunk
exit
!
interface FastEthernet0/3
  switchport trunk encapsulation dot1q
  switchport mode trunk
exit
!
do write
```

7.5 Distribution Switch 1 (VLAN 10 & 20)

Dist-Switch-1

```
enable
configure terminal
hostname Dist-Switch-1
!
vlan 10
  name Lab3
exit
vlan 20
  name TC
exit
!
interface FastEthernet0/1
  switchport trunk encapsulation dot1q
  switchport mode trunk
exit
```

```
!  
interface FastEthernet0/2  
  switchport trunk encapsulation dot1q  
  switchport mode trunk  
exit  
!  
interface FastEthernet0/3  
  switchport trunk encapsulation dot1q  
  switchport mode trunk  
exit  
!  
do write
```

7.6 Distribution Switch 2 (VLAN 30 & 40)

Dist-Switch-2

```
enable  
configure terminal  
hostname Dist-Switch-2  
!  
vlan 30  
  name Lab1  
exit  
vlan 40  
  name Lab2  
exit  
!  
interface FastEthernet0/1  
  switchport trunk encapsulation dot1q  
  switchport mode trunk  
exit  
!  
interface FastEthernet0/2  
  switchport trunk encapsulation dot1q  
  switchport mode trunk  
exit  
!  
interface FastEthernet0/3  
  switchport trunk encapsulation dot1q  
  switchport mode trunk  
exit  
!  
do write
```

7.7 Access Switches — FCAI Branch

Switch-Lab3 / Switch-TC / Switch-Lab1 / Switch-Lab2

```
! — Switch-Lab3 (VLAN 10) —  
enable  
configure terminal  
hostname Switch-Lab3  
vlan 10  
  name Lab3  
exit  
interface FastEthernet0/3
```

```
    switchport mode trunk
exit
interface FastEthernet0/1
    switchport mode access
    switchport access vlan 10
exit
interface FastEthernet0/2
    switchport mode access
    switchport access vlan 10
exit
do write
```

```
! — Switch-TC (VLAN 20) —————
```

```
enable
configure terminal
hostname Switch-TC
vlan 20
    name TC
exit
interface FastEthernet0/1
    switchport mode trunk
exit
interface FastEthernet0/2
    switchport mode access
    switchport access vlan 20
exit
interface FastEthernet0/3
    switchport mode access
    switchport access vlan 20
exit
do write
```

```
! — Switch-Lab1 (VLAN 30) —————
```

```
enable
configure terminal
hostname Switch-Lab1
vlan 30
    name Lab1
exit
interface FastEthernet0/1
    switchport mode trunk
exit
interface FastEthernet0/2
    switchport mode access
    switchport access vlan 30
exit
interface FastEthernet0/3
    switchport mode access
    switchport access vlan 30
exit
do write
```

```
! — Switch-Lab2 (VLAN 40) —————
```

```
enable
configure terminal
hostname Switch-Lab2
vlan 40
    name Lab2
exit
interface FastEthernet0/1
    switchport mode trunk
```



```
exit
interface FastEthernet0/2
  switchport mode access
  switchport access vlan 40
exit
interface FastEthernet0/3
  switchport mode access
  switchport access vlan 40
exit
do write
```

HQ Branch Configuration

7.8 HQ Router (Cisco 2911)

HQ-Router — DHCP Server + OSPF

```
enable
configure terminal
hostname HQ-Router
!
interface Serial0/3/0
 ip address 192.168.2.2 255.255.255.252
 no shutdown
exit
!
interface GigabitEthernet0/0
 no shutdown
exit
!
interface GigabitEthernet0/1
 no shutdown
exit
!
! VLAN 70 & 80 on Gig0/0
interface GigabitEthernet0/0.70
 encapsulation dot1Q 70
 ip address 192.168.1.193 255.255.255.224
exit
!
interface GigabitEthernet0/0.80
 encapsulation dot1Q 80
 ip address 192.168.1.225 255.255.255.224
exit
!
! VLAN 50 & 60 on Gig0/1
interface GigabitEthernet0/1.50
 encapsulation dot1Q 50
 ip address 192.168.1.129 255.255.255.224
exit
!
interface GigabitEthernet0/1.60
 encapsulation dot1Q 60
 ip address 192.168.1.161 255.255.255.224
exit
!
ip dhcp pool IT-Pool
 network 192.168.1.128 255.255.255.224
 default-router 192.168.1.129
 dns-server 192.168.1.3
exit
!
ip dhcp pool HR-Pool
 network 192.168.1.160 255.255.255.224
 default-router 192.168.1.161
 dns-server 192.168.1.3
exit
!
ip dhcp pool PM-Pool
 network 192.168.1.192 255.255.255.224
```

```
default-router 192.168.1.193
dns-server 192.168.1.3
exit
!
ip dhcp pool TL-Pool
network 192.168.1.224 255.255.255.224
default-router 192.168.1.225
dns-server 192.168.1.3
exit
!
router ospf 1
network 192.168.1.128 0.0.0.255 area 0
network 192.168.2.0 0.0.0.3 area 0
exit
!
do write
```

7.9 Collapsed Core Switch — HQ Projects Building

Collapsed-Core

```
enable
configure terminal
hostname Collapsed-Core
!
vlan 50
name IT
exit
vlan 60
name HR
exit
!
interface FastEthernet0/1
switchport trunk encapsulation dot1q
switchport mode trunk
exit
!
interface FastEthernet0/2
switchport trunk encapsulation dot1q
switchport mode trunk
exit
!
do write
```

7.10 Switch-IT-HR

Switch-IT-HR

```
enable
configure terminal
hostname Switch-IT-HR
!
vlan 50
name IT
exit
vlan 60
name HR
```

```
exit
!  
interface FastEthernet0/5  
  switchport mode trunk  
exit  
!  
interface FastEthernet0/1  
  switchport mode access  
  switchport access vlan 50  
exit  
interface FastEthernet0/2  
  switchport mode access  
  switchport access vlan 50  
exit  
!  
interface FastEthernet0/3  
  switchport mode access  
  switchport access vlan 60  
exit  
interface FastEthernet0/4  
  switchport mode access  
  switchport access vlan 60  
exit  
!  
do write
```

7.11 Switch-PM-TL

Switch-PM-TL

```
enable  
configure terminal  
hostname Switch-PM-TL  
!  
vlan 70  
  name PM  
exit  
vlan 80  
  name TL  
exit  
!  
interface FastEthernet0/3  
  switchport mode trunk  
exit  
!  
interface FastEthernet0/1  
  switchport mode access  
  switchport access vlan 70  
exit  
!  
interface FastEthernet0/2  
  switchport mode access  
  switchport access vlan 80  
exit  
!  
do write
```

WAN Interconnection — OSPF Routing

8. WAN Link and OSPF Summary

The two branches are connected via a Serial WAN link with OSPF Area 0 configured on both routers.

Router	Serial Interface	IP Address	OSPF Networks
FCAI-Router	Serial0/3/0	192.168.2.1/30	192.168.1.0/24, 192.168.2.0/30
HQ-Router	Serial0/3/0	192.168.2.2/30	192.168.1.128/24, 192.168.2.0/30

OSPF ensures dynamic route exchange so all VLANs in both branches can communicate across the WAN link without static routes.

End of Report